

SIGHT Project - Meeting 25

Project Name: Safety Immersion and Gamified Hazard Training for Industry 5.0 Workers

Project Number: WSIC26-250206-009

Date: September 04, 2026

Time: 11:00 AM EDT

Location: Zoom

Meeting Focus: Administrative updates, new team member introductions, WebVR milestone tracking, AR SOPs, MaxByte's annotation progress, and a closed session on immersive environment testing and AR status.

Attendees

- **Miami University (MU):** Fadel Megahed (PI), Arthur Carvalho (Co-PI), Mohammad Mayyas (Co-PI), Jay Shan (Co-PI), Ibrahim Yousif (Associate Research Scientist), Michael Wise (GRA), Austin Hamilton (GRA/SIGHT AR Lead), Wissam Nusair (Student Research Assistant), Mo Farrag (Project Manager)
- **Discovery Center:** Kristen Morio (Evaluator), Yue Li (Evaluator), Abby Helsing (Senior Research Associate/Evaluator)
- **External Consultant:** Lora Cavuoto (University of Buffalo)
- **MaxByte:** Ramshankar C S (CEO)
- **Industrious Group Inc.:** Jeff Malek (Corporate Director of EHS)

1. Introductory Remarks & Recap / Approval of Minutes

- **Welcome & Recap:** Fadel Megahed opened the meeting by thanking the team for their aggressive pace and excellent work over the summer.
- **Approval of Minutes:** The minutes for Meeting 24 were circulated previously. With no feedback or corrections received, they are considered officially approved.

2. Administrative Updates

- **BWC Deliverables Status:**
 - The Year 1 interim report and Q4 invoice were accepted and processed, and funds have been reimbursed.

- The Q5 report is due October 9, 2026. The MU team aims to have a draft ready by October 2.
- The BWC Q5 presentation and attendance verification are scheduled for October 21/22, 2026.
- **New Team Members:**
 - **Lindsay Stevenson (MSBA):** Joining as a Student Research Assistant to support project analytics and AI updates (absent due to a class conflict).
 - **Wissam Nusair (Freshman):** Joining as a Student Research Assistant. Wissam comes with an impressive background (internships at P&G and Fifth Third) and is supporting the digital twin and AM Hub operations.
 - **Abby Helsingier:** Joining the Discovery Center evaluation team. Abby brings 15 years of experience in public health and serves as a Senior Research Associate.
- **ASSP Safety 2027 Conference:** Fadel and Lora are finalizing an abstract for the ASSP Safety 2027 conference in New Orleans focusing on how generative AI can inform safety training materials.

3. Technical Updates: VR Updates and Updated SOPs for the AR

- **WebVR Progress (Arthur):**
 - Arthur tracked 85 issues via GitHub, grouping them into milestones (CNC, Milling, Cobot, Dashboard, AI). The team successfully addressed a sensitive unencrypted network connection bug (identified by Austin).
 - *Next Steps:* Revamping the evaluation module. Currently, assessments feel too repetitive (mirroring the training). Arthur plans to implement distinct evaluation methods (e.g., multiple-choice questions and error identification on an avatar) for version 2.0.
- **Machine SOPs for AR (Michael):**
 - Michael presented the finalized SOPs for the CNC Lathe and UR5e Cobot. These documents have been translated into an AR-ready structure that highlights exactly which components (e.g., teach pendant, e-stop) need to be annotated in the AR headset.
 - Established clear "safety zones" (e.g., a 4-foot barrier around the Cobot) to be integrated into the headset's spatial awareness.
- **Digital Twin Demo (Wissam):** Wissam demonstrated the centralized digital twin website, which allows users to view 2D/3D layouts of the AM Hub, access live telemetry data from machines like the Cobot, and interact with the equipment remotely.

4. Industry Partner Collaboration

- **MaxByte Update (Ramshankar):**
 - MaxByte is progressing on dataset annotations and work instructions for the AR environment.

They have captured thousands of images of the CNC and Cobot operations.

- *Feedback:* Ibrahim requested that MaxByte migrate the CNC annotations from Google Drive to RoboFlow to ensure all data remains centralized.
- *Note on Documentation:* Fadel requested that future documents submitted for review should list "Miami Approval" as a unified team rather than requiring individual sign-offs from specific MU members.

(Note: Industry partners Jeff Malek and Ramshankar C S as well as the Discovery Center Evaluation team, left the meeting prior to the closed session per the agenda).

5. Miami-Lora Closed Session: Immersive Environment Testing

- **AR Environment Status (Austin):**
 - Austin showcased live on-device (Magic Leap 2) footage of the AR environment.
 - The system now connects to the live machine state via WebSockets.
 - *New Features:* Added step numbering to instructions and implemented hand gestures (e.g., thumbs-up/down for "next/previous") to avoid the inaccuracies of hand-tracking UI clicks.
 - *Anchor Drift Mitigation:* Created a supplemental visual tag system to quickly realign UI panels when physical components (like the CNC control panel) move.
- **Hardware Risk Assessment (Magic Leap 2):**
 - The team noted that the Magic Leap 2 headset has been discontinued.
 - To mitigate the risk of catastrophic hardware failure, Jay will coordinate with the department to purchase a backup headset via eBay using Miami funds (non-BWC funds).
 - Austin and the AM Hub subteam will immediately begin storing the current headset in a locked cabinet in Ibrahim's office.
- **Experimental Protocol:** The discussion on human subject testing design (usability, perceived usefulness, and performance) has been tabled for a dedicated breakout meeting next week.

6. Risk Mitigation and Log Updates

- **Hardware Obsolescence:** The discontinuation of the Magic Leap 2 was formally added to the risk log. While a backup device is being procured for this Track 2 project, the team will need to evaluate alternative industry-standard hardware (e.g., Meta Quest) for future commercialization and Track 3 grant applications.

7. Further Discussion and Wrap-Up

- Fadel will schedule a meeting for Friday at 11:10 AM EDT to review the immersive environment testing protocol with Lora.
- MU team research leads will meet with the BWC on Tuesday at 9:30 AM to discuss Phase 3 timelines and personnel planning.
- Fadel will provide the two slide decks from today's meeting to the BWC.

Action Items

Task	Responsible	Deadline
Move CNC annotations from Google Drive to RoboFlow	Ramshankar / MaxByte	ASAP
Procure backup Magic Leap 2 headset (eBay) via MU P-Card	Jay Shan / Mohammad Mayyas	ASAP
Secure the current Magic Leap 2 headset in Ibrahim's locked office	Ibrahim Yousif / Austin Hamilton	Immediate
Prepare and circulate the Q5 Report team draft	Fadel Megahed	October 2, 2026
Schedule a follow-up meeting to discuss the human subject testing protocol	Fadel Megahed	Next Friday (12:45 PM)
Send weekly summary of AR developments to Jay	Austin & Michael	Weekly
Assist Austin with CV model integration for the CNC CAD/3D	Ibrahim Yousif	Next Phase
Provide Meeting 25 slide decks to the BWC	Fadel Megahed	ASAP